Project Design Phase-II

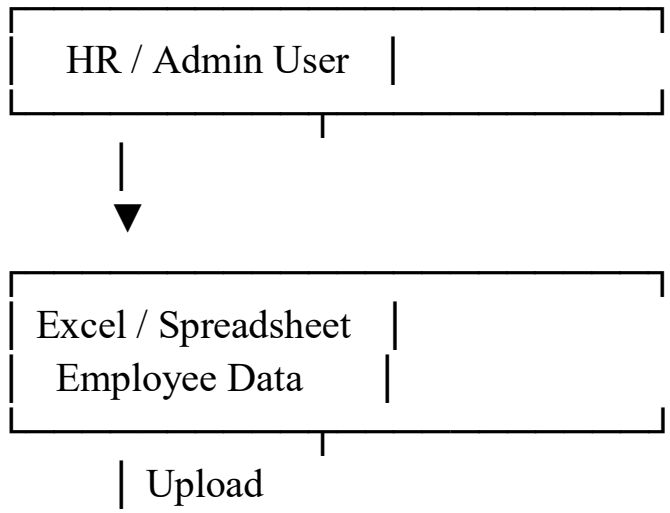
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	6ab24e5e6372d772b2c88edb
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture

The proposed solution uses the **ServiceNow platform** to import employee data from an Excel spreadsheet. The spreadsheet data is first loaded into an **Import Set table**. A **Transform Map** maps the source fields to the corresponding fields in the **Employee Test** target table. **Coalesce** is used to identify existing employee records and prevent duplicate records. The transformed data is then validated using **Transform History** and displayed through **Reports and Employee Analytics Dashboard**.

Architecture Flow



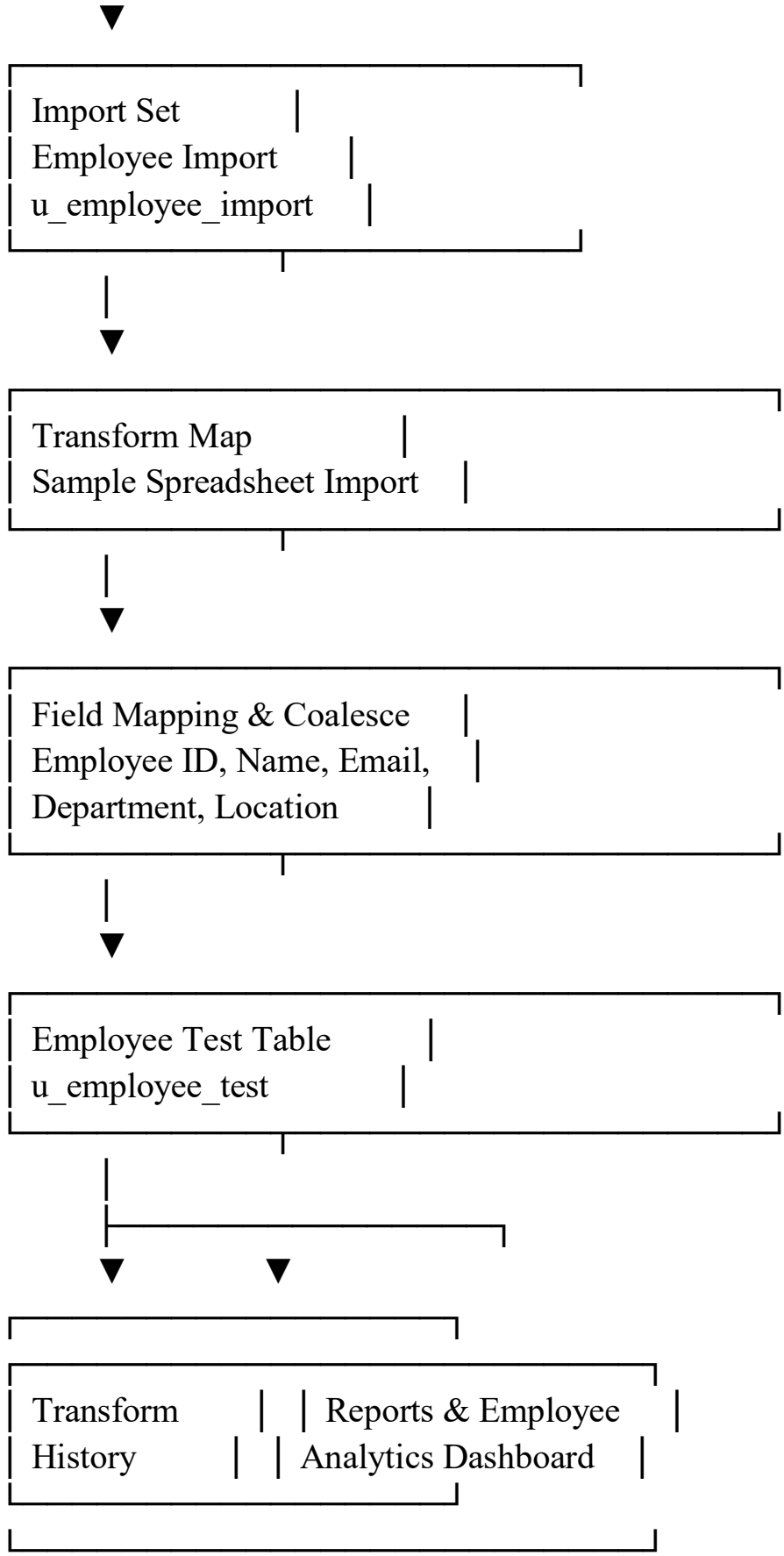

Import Set
Employee Import
u_employee_import
Transform Map
Sample Spreadsheet Import
Field Mapping & Coalesce
Employee ID, Name, Email,
Department, Location
Employee Test Table
u_employee_test
Transform
History
Reports & Employee
Analytics Dashboard

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	**User Interface**	Interface through which the HR/Admin user uploads the employee spreadsheet, manages imported data and views reports and dashboards.	ServiceNow Web UI
2	**Application Logic-1**	Handles loading of employee data from the spreadsheet into the staging Import Set table.	ServiceNow Import Sets
3	**Application Logic-2**	Transfers and transforms employee data from the Import Set table to the Employee Test target table.	ServiceNow Transform Maps
4	**Application Logic-3**	Maps employee fields and identifies existing records to prevent duplicate employee records.	Field Mapping and Coalesce
5	**Database**	Stores the imported employee data and transformed employee records in ServiceNow tables.	ServiceNow Database
6	**Cloud Database**	Not separately used in the proposed solution. Employee data is stored within the ServiceNow platform.	ServiceNow Platform
7	**File Storage**	Stores the prepared employee spreadsheet file used for importing employee data.	Excel / Spreadsheet File
8	**External API-1**	No external API is required for importing and transforming employee spreadsheet data.	Not Applicable

9	**External API-2**	No external API is required for employee data transformation and validation.	Not Applicable
10	**Machine Learning Model**	No machine learning model is required. The solution uses rule-based data import, field mapping and transformation.	Not Applicable
11	**Infrastructure (Server / Cloud)**	The application is developed and executed on the ServiceNow platform using its web-based cloud environment.	ServiceNow Cloud Platform

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Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	No separate open-source framework is required. The solution uses the built-in ServiceNow platform features for data import, transformation, reporting and dashboard creation.	ServiceNow Platform
2	**Security Implementations**	Access to employee data and application features is controlled through ServiceNow user roles and access controls. Employee data is managed within the ServiceNow platform.	ServiceNow Roles and Access Controls

3	**Scalable Architecture**	The solution can handle increasing employee records by using Import Sets and Transform Maps for repeated spreadsheet imports. The target table can store additional employee records as required.	ServiceNow Import Sets and Transform Maps
4	**Availability**	The solution is available through the ServiceNow web platform whenever the ServiceNow instance is accessible. Users can perform imports, validation and reporting through the platform.	ServiceNow Cloud Platform
5	**Performance**	Import Sets provide a structured staging process, while Transform Maps automate field mapping and data transformation. Coalesce helps avoid unnecessary duplicate records during repeated imports.	Import Sets, Transform Maps and Coalesce